

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:50

Annexure A

Pseudocode Writing Task (Algorithm Design)

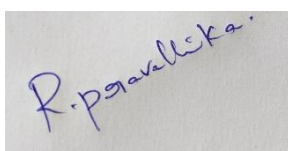
Criteria	Problem Understanding (2)	Algorithm Design (3)	Use of Control Structures (2)	Pseudocode Structure (1)	Completeness (2)	Total(10)
<i>Weightage</i>	20%	30%	20%	10%	20%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I RAJANEDI DATI PRAVALLIKA/(192525473)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
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O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones,
Charger O5	Mobile, Earphones

Question:

Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration. (BL3 – 1.05 Marks)

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
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B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine
Learning B5	Python, Machine Learning

Question:

Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. (BL4 – 1.05 Marks)

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3

{Internet, Streaming} 3
{Streaming, Cloud Storage} 2

Question:

Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. *(BL3 – 1.05 Marks)*

Q4. Dataset: Student Course Enrollment

Student ID Courses Enrolled

S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:

Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course "**Python**" are considered during the mining process. Explain how the constraint affects candidate generation and pruning. *(BL4 – 1.05 Marks)*

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID Treatments Received

P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Question:

Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

1. Apriori Algorithm - online Shopping Transaction.

Given dataset.

Order	Product
1.	Mobile earphones.
2.	Mobile charger
3.	EarPhones, charger
4.	Mobile, earphones.
5.	Mobile earphones.

Assume minimum support count = 3

pseudocode.

APRIORI (Transaction-Dataset, Minsupport)

1. C_1 - generate all itemsets.
2. Count support of every item in C_1 .
3. $C_1 \leftarrow$ remove items whose support $<$ min-support.
4. $k \leftarrow 2$
5. while $L(k-1)$ is not empty :
 $C_k \leftarrow$ generate candidates from $(k-1)$
for each candidate c in C_k :
 If any $(k-1)$ -subset of c is not in $L(k-1)$:

Remove c

Count support of remaining candidates

$L_k \leftarrow$ candidates having support
min-support.

$k \leftarrow k+1$

6. Return all frequency itemsets $L_1, L_2, \dots$

2. FP-growth Library book Borrowing

Dataset.

Borrow ID

Book

B_1

Data science, python

B_2

Python, Machine learning

B_3

Data science, Machine
Learning

B_4

Data science, python,
Machine Learning

B_5

python, Machine learning

Minimum support count = 2.

Pseudocode

FP-growth (D , min-support)

1. Scan D and calculate support of each.

item.

3. sort remaining items according to decreasing support.

4. Create an empty F-p Tree.

5. For each transaction T in D :

Remove increment it
update node count.

6. Create header table linking identical item.

7. Mine the Fp-Tree recursively

Find conditional pattern bases.

Construct conditional Fp-trees.

Generate frequent item sets.

8. Return all frequent itemsets.

3. Constraint -Based frequent itemset Mining.

Dataset

student

S_1

S_2

S_3

S_4

S_5

courses

Python, Database

Python, AI

Database, AI

Python

Python.

Final frequent itemsets:

$\{ \text{pytho} \}$

$\{ \text{python}, \text{Datasets} \}$

$\{ \text{python}, \text{AI} \}$

Pseudo code

1. Define Constraint
2. Generate C_1
3. Remove Candidate that do not contain python.
4. count Support
5. $L_1 \leftarrow$ candidates Satisfying min-support
6. for $k=2$ while $L(k-1)$ is not empty
Generate candidate itemset from $(k-1)$
For each candidate c :
If "python" is not in c .
 prune c
Else
 Add c to L_k
7. Return all L_k .

Q4. Association Rule Generation

Dataset.

Itemset.

Support count

{Internet}	4
{streaming}	4
{cloud storage}	3
{Internet, streaming}	3
{streaming, cloud, storage}	2

Assume minimum confidence, 60,

$$\text{Confidence } (A \rightarrow B) = \frac{\text{support } (A \cup B)}{\text{support } (A)} \times 100$$

Pseudo code

1. For each frequent itemset
2. Generate all non-empty subsets A of F
3. $B \leftarrow F - A$
4. $\text{confidence} \leftarrow \text{support}(A) / \text{support}(A \cup B)$
5. If confidence $\geq$ min-confidence.
6. Generate rule $A \rightarrow B$
7. Accept the rule.
8. Else:
9. Reject the rule.
10. RETURN all accepted rules.

5. Multilevel Association Rule Mining.

Dataset

Patient

treatments

P₁

Healthcare, Cardiology

P₂

ECG,

P₃

Health care, Orthopedics

P₄

Healthcare, Cardiology

P₅

Healthcare.

Healthcare

Pseudocode:-

1. Define the treatment hierarchy
2. start from highest / general level
3. For each hierarchy level.
Extract items belonging to that level
count support of each min-support
4. Move to the next, more detailed
hierarchy level.
5. Repeat the mining process
6. RETURN frequent patterns &
rule from all levels.